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REIN: Residual Modulation of Time Series Forecasts via LLM-Refined News Regime Signals

(/pdf?id=moaHRVEI44)

Zhaoge Bi (/profile?id=~Zhaoge_Bi1),
Linghan Huang (/profile?id=~Linghan_Huang1),
Jiawen Wen (/profile?id=~Jiawen_Wen1),
Qingwen Zeng (/profile?id=~Qingwen_Zeng1),
Huaming Chen (/profile?id=~Huaming_Chen1)

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TL;DR: REIN: a two-stage framework that uses LLM-refined news as a bounded modulator of a structured residual (slow / shape / spike) on top of a frozen time-series base, with a counterfactual protocol attributing gains to news content.

Abstract:
Time series forecasts in operational domains are often disrupted by external events such as heatwaves, generator outages, policy interventions, supply shocks, and market incidents. Most deep forecasters rely on numerical histories, while recent text-augmented methods offer limited control over how far text can move a forecast, which temporal component text should influence, and whether observed gains reflect time-aligned news content rather than news availability or calendar correlations. We introduce REIN (Residual Information from News), a two-stage framework that reformulates news-aware forecasting as bounded residual modulation. An LLM agent distills raw news records into a leakage-controlled regime bank of compact, time-aligned signals. A PatchTST-based delta module decomposes the residual of a frozen base forecaster into persistent-level, intra-day-shape, and extreme-event-spike channels. News does not write forecasts directly. It only scales these residual channels through a small set of bounded modulation knobs. This construction enforces a bound on deviation from the base forecast and provides a clean reversion path when the news bank is empty. A counterfactual protocol with news blanking and date permutation, evaluated on disjoint news-active and news-inactive subsets, provides sample-level evidence that improvements are tied to selective use of aligned news. On four heavy-tailed real-world datasets (Electricity Demand, Electricity Price, Traffic, Gas Demand) at horizons up to 720 steps, REIN improves MAE by 2--21% over a strong DLinear base and is the best or second-best model on most evaluated settings. Blanking the news increases news-active MAE on all four datasets, by up to 42.6% on the most news-sensitive setting (Traffic, $H=96$), supporting the view that REIN benefits from structured news signals rather than from adding text indiscriminately.

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Revisions (/revisions?id=gmHCfi3oa2)

Metareview:

Summary: In this paper, the authors propose REIN, a two-stage framework for news-aware time-series forecasting. Specifically, a numerical base forecaster is trained on historical data and frozen. Then, an LLM agent distills raw news into a "regime bank" of compact signals (actionability, topic, intensity, duration, confidence). These signals modulate three structured residual channels, including persistent level, intra-day shape, and extreme-event spike, via bounded modulation knobs on top of the frozen base forecast. The bounded design ensures that news influence is capped and reverts to the base forecast when no actionable news is present. Evaluation on four datasets (electricity price, demand, gas demand, traffic) uses counterfactual blanking and date permutation to test whether gains stem from aligned news content.

Main strengths:

1. The paper identifies a real limitation, which is standard forecasting models rely on numerical histories and cannot easily incorporate contemporaneous external events. This motivates the work very well.
2. Capping news influence and reverting to a reliable numerical baseline when no actionable news exists is a practical safeguard against noisy or irrelevant text. This enables sensible bounded modulation design.
3. This works also compares activate, blanked, and date-permuted regime packs, which provides evidence that improvements are tied to aligned news content rather than mere presence of text features.

Main weaknesses:

1. Based on Reviewer czkv and Reviewer uvzL, REIN is compared against numerical-only baselines but does not include text-augmented methods such as ChatTime or Time-VLM. Since REIN has access to extra time-aligned news, the comparison is not fair. The paper's stated contribution is in the way news is integrated, but without news-aware baselines, it is unclear whether the gains come from the architecture or simply from having access to additional information.
2. The paper introduces multiple design choices, e.g., five-axis regime vector, active threshold, confidence-weighted aggregation, slow/shape/spike decomposition, but does not systematically ablate them. It is unclear which components are necessary. Reviewer uvzL explicitly requests ablation of the three residual channels and Reviewer czkv asks for ablations of the main design choices.
3. REIN fails to outperform numerical baselines on Traffic. Table 3 shows only marginal improvements over the permuted regime pack in many cases, raising questions about whether gains are genuinely due to aligned content rather than mere presence of features.
4. Also according to Reviewer czkv, though this paper claims that slow, shape, and spike channels correspond to meaningful temporal effects, but no ground-truth labels, human evaluation, or synthetic experiments are provided to validate this.
5. Experiments rely on dataset-specific news sources, prompts, thresholds, and modulation caps. It remains unclear whether REIN transfers to new domains or noisier corpora without substantial re-tuning.

Given the above weakness, the required changes are substantial.

1. The authors must compare REIN against at least one strong text-augmented baseline (ChatTime, Time-VLM, or similar) under the same news corpus. Without this, the evaluation is fundamentally unfair.

2. Systematic ablations of the five-axis regime vector, active threshold, confidence weighting, and slow/shape/spike decomposition are needed to demonstrate that each component contributes meaningfully.
3. Also, a synthetic experiment where known level shifts, shape changes, or spikes are induced, with ground-truth comparison against learned residual channels, would substantiate the interpretability claim.
4. More datasets, domains, or noisier corpora would clarify generalizability substantially.

Currently, without such changes, the evaluation remains insufficient to support the claims.

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We thank the AC and reviewers for their constructive and detailed feedback. In response, we have strengthened both the empirical evaluation and the methodological clarification of REIN.

First, we added two text-augmented baseline using ChatTime and Time-VLM. The results show that REIN outperforms both of the baselines in most completed settings. This supports that REIN's gains are not simply due to access to additional news information, but come from its controlled residual-modulation mechanism. ChatTime experiments for horizons 192 and above are still running due to its substantially longer runtime.

Second, we added a synthetic component-validation experiment on the NSW Electricity Price dataset at horizon 48. By injecting known level shifts, shape distortions, localized

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Summary:
This paper proposes REIN, a two-stage framework that reformulates news-aware forecasting as bounded residual modulation. In the base stage, a numerical forecaster is trained on historical time-series data and then froze. In the delta stage, an LLM agent distills raw news records into a leakage-controlled regime bank of compact, time-aligned signals, including actionability, topic, intensity, duration, and confidence. These regime signals modulate structured residual channels on top of the frozen base forecast. Specifically, REIN decomposes the residual into persistent-level, intra-day-shape, and extreme-event-spike channels, and scales them through bounded modulation knobs. The paper evaluates REIN on four real-world datasets and uses news blanking and date permutation to test whether improvements are tied to aligned news content.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths:

1. The paper identifies a real limitation of many forecasting models: they mainly rely on numerical histories and cannot easily use contemporaneous external events before those events are reflected in the time series.
2. The bounded residual modulation design is a reasonable way to control text influence. By letting news only modulate residual channels rather than directly generate forecasts, REIN reduces the risk that noisy or irrelevant news arbitrarily changes the prediction.
3. The counterfactual protocol is a useful evaluation idea. Comparing active, blanked, and date-permuted regime packs provides some evidence on whether the model benefits from aligned news content rather than simply from the existence of text features.

Weaknesses:

1. The designs of the news regime bank and delta module are mostly engineering-oriented. Many choices, such as the five-axis regime vector, confidence-weighted aggregation, active threshold, slow/shape/spike decomposition, and bounded modulation knobs, are insufficiently justified. The lack of systematic ablations makes it unclear which components are truly necessary.
2. The evaluation does not fully support the effectiveness and generalizability of REIN. The baselines mostly use only numerical histories and lack text-augmented methods, such as ChatTime [1] and Time-VLM [2]. Since REIN has access to extra time-aligned news, the comparison is not completely fair. Moreover, the gains are not consistently strong: REIN underperforms on Traffic in Table 1, and Table 3 shows only marginal improvements over the permuted regime pack in many cases. The evaluation is also limited to only four datasets, which is insufficient to support broad generalization claims. In addition, the experiments rely on dataset-specific news sources, prompts, thresholds, and modulation caps, so it remains unclear whether REIN can transfer to new domains or noisier news corpora without substantial re-tuning.
3. The component-level interpretability claim is not sufficiently validated. REIN decomposes residuals into slow, shape, and spike channels, but does not provide ground-truth labels, human evaluation, or quantitative evidence showing that these components are meaningful.

[1] Wang et al. Chattime: A unified multimodal time series foundation model bridging numerical and textual data. AAAI 2025.

[2] Zhong et al. Time-VLM: Exploring Multimodal Vision-Language Models for Augmented Time Series Forecasting. ICML 2025.

Quality: 2: not good
Clarity: 3: good
Significance: 2: not good
Originality: 2: not good
Questions:

1. Can the authors provide stronger justification and ablations for the main design choices in the news regime bank and delta module? For example, what happens if the five-axis regime vector, active threshold, confidence weighting, or slow/shape/spike decomposition is removed or replaced?
2. Can the authors compare REIN with stronger text-augmented or news-aware forecasting baselines under the same news corpus? Results on more datasets, domains, or noisier corpora would also help clarify whether REIN generalizes beyond the current setting.
3. Can the authors provide evidence that the slow, shape, and spike channels correspond to meaningful temporal/event effects? A useful validation could be a synthetic experiment where news events are designed to induce known level shifts, shape changes, or spikes, so the learned residual channels can be compared against component-level ground truth.

Limitations:
The paper mentions that REIN works best when news has a direct effect on the target and may struggle with rare large spikes. However, it does not fully address key limitations, such as the lack of text-augmented forecasting baselines.

Rating: 2: Reject: For instance, a paper with technical flaws, weak evaluation, inadequate reproducibility and incompletely addressed ethical considerations.
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Component-wise ablation results (active threshold, slow/shape/spike)

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Official Comment by Authors 30 Jul 2026, 23:18

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Comment:
Following the previous response, we attach the full component-wise ablation results. All values are rounded to two decimal places, with large values reported in scientific notation for readability. We consider four ablated variants: removing the active threshold, slow component, shape component, and spike component. These variants test whether low-confidence news activations should be filtered, and whether persistent shifts, trajectory-level changes, and abrupt event-driven deviations are all necessary.

The active threshold improves robustness under noisy news inputs by filtering weak regime activations. The slow and shape components are especially important in medium- and long-horizon settings, particularly for NSW traffic and Netherlands gas demand, indicating that news affects both persistent regime trends and future trajectory patterns. The spike component is more dataset- and horizon-dependent, but remains useful for capturing short-term event-driven fluctuations. These results also support the use of bounded modulation knobs and confidence-weighted aggregation. Several ablations lead to unstable increases in MSE on high-variance datasets, suggesting that news signals should modulate the temporal backbone in a controlled manner rather than as unconstrained additive features. Confidence weighting further helps suppress weak or ambiguous news signals while preserving reliable regime evidence.

Overall, the full model REIN benefits from all ablated variants across the four datasets and five prediction horizons. These results show that the proposed components may not be equally important across different datasets, but they are functionally necessary for different forecasting conditions. The ablation study therefore

provides systematic evidence that the news regime bank and delta module are not only engineering-oriented design choices, but targeted mechanisms that improve robustness and expressiveness in news-informed time-series forecasting.

Dataset	Horizon	w/o active threshold MSE	w/o active threshold MAE	w/o slow MSE	w/o slow MAE	w/o shape MSE	w/o shape MAE	w/o spike MSE	w/o spike MAE
Electricity Price	48	560,685.10	106.24	560,039.55	105.38	561,424.12	106.11	553,827.69	99.19
Electricity Price	96	580,703.41	120.88	581,329.79	120.99	579,985.25	117.95	570,403.35	110.54
Electricity Price	192	589,744.70	114.57	584,344.97	110.13	590,413.45	112.80	584,993.64	104.93
Electricity Price	336	625,885.19	124.93	623,768.32	124.14	619,604.39	119.84	616,840.37	114.58
Electricity Price	720	698,547.88	114.06	698,021.48	113.89	696,125.37	109.28	696,719.73	113.07
Electricity Demand	48	494,654.46	458.40	521,448.48	482.61	493,922.61	476.08	519,490.11	474.67
Electricity Demand	96	647,884.88	550.22	644,499.42	550.40	653,781.43	554.44	661,807.70	557.41
Electricity Demand	192	804,204.62	623.19	816,046.95	630.06	814,477.24	637.86	769,887.03	605.76
Electricity Demand	336	956,976.66	687.11	954,247.39	691.22	963,248.46	696.70	930,989.37	678.36
Electricity Demand	720	1,054,914.88	737.53	1,039,622.82	713.62	1,059,919.15	733.79	1,016,792.67	704.30
Traffic	48	3.18E+08	1.16E+04	3.56E+08	1.26E+04	3.06E+08	1.19E+04	3.52E+08	1.22E+04
Traffic	96	3.62E+08	1.28E+04	4.05E+08	1.38E+04	3.66E+08	1.34E+04	3.86E+08	1.37E+04
Traffic	192	4.30E+08	1.38E+04	4.41E+08	1.45E+04	4.44E+08	1.45E+04	4.48E+08	1.46E+04
Traffic	336	4.02E+08	1.40E+04	4.65E+08	1.56E+04	4.45E+08	1.47E+04	5.61E+08	1.67E+04
Traffic	720	7.24E+08	1.90E+04	8.51E+08	2.18E+04	7.62E+08	1.94E+04	7.41E+08	1.97E+04
Gas Demand	48	2.88E+13	4.29E+06	2.91E+13	4.21E+06	3.02E+13	4.28E+06	2.96E+13	4.25E+06
Gas Demand	96	3.69E+13	5.02E+06	3.47E+13	4.79E+06	3.75E+13	4.99E+06	3.64E+13	4.92E+06
Gas Demand	192	3.90E+13	5.22E+06	3.86E+13	5.21E+06	4.02E+13	5.21E+06	3.84E+13	5.14E+06
Gas Demand	336	4.27E+13	5.36E+06	4.08E+13	5.32E+06	4.36E+13	5.38E+06	4.18E+13	5.31E+06
Gas Demand	720	5.57E+13	6.13E+06	5.51E+13	6.09E+06	5.55E+13	6.11E+06	5.36E+13	6.03E+06

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New results for text-augmented baselines

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30 Jul 2026, 23:12

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer czkv

Comment:

In addition to the ongoing ChatTime experiments, we also evaluated Time-VLM on the same four datasets and horizons. Values are rounded to two decimals, with large values in scientific notation. We used long-term forecasting with a 48-point history and prediction lengths of 48, 96, 192, 336, and 720. Inputs are univariate and metrics are reported on the original scale. Time-VLM uses its standard text interface: a fixed dataset-level prompt/domain description with the image-transformed numerical series. It does not ingest sample-specific, time-aligned news. In contrast, REIN uses temporally aligned news summaries and impact/relevance information to modulate forecasting representations. This comparison supports REIN while showing that gains are not uniformly large. Across 20 dataset-horizon combinations, REIN achieves lower MAE in all 20 cases and lower MSE in 19 cases. On average, it reduces MSE by about 9.8% and MAE by about 10.3%. Improvements are clearer on Electricity Demand, Gas Demand, and most Traffic horizons; Electricity Price shows smaller MSE gains but clearer MAE gains. The only exception is Traffic at horizon 720, where REIN has worse MSE but slightly better MAE, suggesting sensitivity to tail errors. We therefore do not claim REIN is universally dominant; rather, the results show that time-aligned news conditioning is generally beneficial relative to Time-VLM, with dataset- and horizon-dependent trade-offs.

Dataset	Horizon	Time-VLM's MSE	Time-VLM's MAE	REIN's MSE	REIN's MAE
Electricity price	48	590,164.75	133.11	561,709.83	107.32
Electricity price	96	606,456.50	134.22	580,792.24	120.36
Electricity price	192	605,853.25	128.67	587,785.51	110.56
Electricity price	336	632,645.50	134.42	622,154.88	123.54
Electricity price	720	718,606.88	147.68	697,551.00	112.20
Electricity demand	48	562,758.69	512.11	503,629.23	469.81
Electricity demand	96	775,437.94	613.38	658,304.61	554.13
Electricity demand	192	916,833.75	677.49	803,765.76	620.65
Electricity demand	336	1,042,423.69	724.26	969,312.19	695.26
Electricity demand	720	1,146,111.63	768.34	1,044,934.50	722.82
Traffic	48	4.18E+08	1.52E+04	3.27E+08	1.20E+04
Traffic	96	4.91E+08	1.64E+04	3.65E+08	1.31E+04
Traffic	192	4.34E+08	1.53E+04	3.96E+08	1.35E+04
Traffic	336	4.47E+08	1.58E+04	4.08E+08	1.41E+04
Traffic	720	6.14E+08	1.87E+04	6.74E+08	1.84E+04
Gas Demand	48	3.56E+13	4.55E+06	3.02152E+13	4.28E+06
Gas Demand	96	4.57E+13	5.42E+06	3.74725E+13	5.00E+06
Gas Demand	192	4.74E+13	5.58E+06	3.98071E+13	5.20E+06
Gas Demand	336	5.03E+13	5.72E+06	4.35579E+13	5.37E+06
Gas Demand	720	5.99E+13	6.25E+06	5.49972E+13	6.09E+06

We also updated the ChatTime comparison on the same REIN datasets. After additional effort, we were able to make ChatTime run at horizon 336, but that experiment has not finished yet. REIN becomes better at the current three horizons, except on Electricity Price at horizon 48, ChatTime obtains slightly lower MSE/MAE. ChatTime remains competitive on short-horizon Electricity Price. Therefore it is necessary to present these results as evidence that different text-augmented forecasting approaches have complementary strengths and limitations. We will include these results in the revised manuscript.

"-" indicates that the corresponding experiment has not yet been completed.

Dataset	Horizon	ChatTime's MSE	ChatTime's MAE	REIN's MSE	REIN's MAE
Electricity price	48	564044.4824	103.4364719	561709.8272	107.3152611
Electricity price	96	885512.6037	142.81023	580792.2357	120.3588648
Electricity price	192	804647.0945	143.6059037	587785.5117	110.5641795
Electricity price	336	-	-	-	-
Electricity price	720	-	-	-	-
Electricity demand	48	1503324.343	889.2976611	503629.2308	469.8057475
Electricity demand	96	2019168.523	1079.905995	658304.6118	554.1322286
Electricity demand	192	2255421.184	1157.77084	803765.7583	620.6480936
Electricity demand	336	-	-	-	-
Electricity demand	720	-	-	-	-
Traffic	48	1003876539	22486.90546	327345499.6	12036.87399
Traffic	96	1944465970	31551.93068	365076237.5	13098.16765
Traffic	192	2861108888	38966.76309	396075842	13474.39833
Traffic	336	-	-	-	-
Traffic	720	-	-	-	-
Gas Demand	48	5.83E+13	5.86E+06	3.02152E+13	4.28E+06
Gas Demand	96	6.84E+13	6.55E+06	3.74725E+13	5.00E+06
Gas Demand	192	6.75E+13	6.44E+06	3.98071E+13	5.20E+06
Gas Demand	336	-	-	-	-
Gas Demand	720	-	-	-	-

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Rebuttal by Authors

Rebuttal

by Authors (Qingwen Zeng (/profile?id=~Qingwen_Zeng1), Huaming Chen (/profile?id=~Huaming_Chen1), Zhaoge Bi (/profile?id=~Zhaoge_Bi1), Linghan Huang (/profile?id=~Linghan_Huang1), +1 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission16816/Authors))

28 Jul 2026, 21:58 (modified: 28 Jul 2026, 23:35)

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Revisions (/revisions?id=i2pkZv9nPv)

Rebuttal:

We appreciate the reviewer's careful reading and valuable suggestions, which helped us improve both the empirical analysis and the presentation of the paper.

Response to Q1

The slow–shape–spike design in REIN synthesises ideas from interpretable decomposition, multiscale forecasting, and extreme-event modelling rather than following any single prior method. The slow component is inspired by trend modelling and trend–remainder decomposition in N-BEATS, Autoformer, and DLinear [1]–[3]. The shape component follows multiscale methods such as N-HiTS and TimeMixer, which distinguish coarse global variation from fine local patterns [4], [5]. The spike component is motivated by extreme-event forecasting, where rare observations are handled separately through specialised predictors, classifiers, or gates [6], [7]. REIN combines these ideas at the residual level: after the frozen numerical backbone captures routine dynamics, the remaining error is decomposed into a persistent slow shift, a zero-mean shape adjustment, and a sparse gated spike correction.

We also agree that these design choices should be supported empirically in addition to being motivated architecturally. Hence, we have added ablation studies that remove or weaken each of these components while keeping the same numerical backbone, training protocol, and retrieved news corpus. We will include these results in the comment area later. A clearer discussion of the role of each component will be added in the revision for the empirical validation.

References

[1] B. N. Oreshkin et al., “N-BEATS: Neural Basis Expansion Analysis for Interpretable Time Series Forecasting,” ICLR, 2020.

[2] H. Wu et al., “Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting,” NeurIPS, 2021.

[3] A. Zeng et al., “Are Transformers Effective for Time Series Forecasting?” AAAI, 2023.

[4] C. Challu et al., “N-HiTS: Neural Hierarchical Interpolation for Time Series Forecasting,” AAAI, 2023.

[5] S. Wang et al., “TimeMixer: Decomposable Multiscale Mixing for Time Series Forecasting,” ICLR, 2024.

[6] D. Ding et al., “Modeling Extreme Events in Time Series Prediction,” KDD, 2019.

[7] Y. Li, J. Xu, and D. C. Anastasiu, “An Extreme-Adaptive Time Series Prediction Model Based on Probability-Enhanced LSTM Neural Networks,” AAAI, 2023.

Response to Q2

Thank you for pointing out this for us. We have now included ChatTime as a representative text-augmented time-series forecasting baseline using the same retrieved news corpus. The results in the table below show that REIN outperforms ChatTime in the majority of experimental settings, which supports that the gains of REIN are not merely due to giving the model access to additional textual information. We have not yet completed the ChatTime experiments for prediction horizons of 192 and above, as its runtime is substantially longer than that of REIN under our experimental setting. We will continue running these experiments and include the results once they are available. Due to GPU memory constraints on our RTX 5090 experimental platform, we were not able to successfully run the smoke test for ChatTime for the longer prediction horizons of 336 and 720. Although this limits the comparison at long horizons, it also highlights an important practical advantage of REIN: its news integration mechanism introduces substantially lower computational overhead than ChatTime, making it more scalable for long-horizon forecasting.

Regarding generalization, we would like to clarify that the incorporated datasets cover different levels of domain specificity and news quality. Electricity demand and price use more curated domain-specific feeds, Traffic uses a broader GDELT-derived news stream, and Gas demand uses a noisier bilingual corpus. We will make this distinction clearer in the revised manuscript and discuss Traffic more explicitly as a setting where broad and weakly local news coverage limits the achievable benefit.

Dataset	Horizon	ChatTime's MSE	ChatTime's MAE	REIN's MSE	REIN's MAE
Electricity price	48	564044.4824	103.4364719	561709.8272	107.3152611
Electricity price	96	885512.6037	142.81023	580792.2357	120.3588648
Electricity demand	48	1503324.343	889.2976611	503629.2308	469.8057475
Electricity demand	96	2019168.523	1079.905995	658304.6118	554.1322286
Traffic	48	1003876539	22486.90546	327345499.6	12036.87399
Traffic	96	1944465970	31551.93068	365076237.5	13098.16765
Gas Demand	48	5.83E+13	5859254.306	3.02152E+13	4278464.786
Gas Demand	96	6.84E+13	6548942.089	3.74725E+13	4997697.726

Response to Q3

To further validate the semantic meaning of the slow/shape/spike decomposition, we conducted a synthetic component-validation experiment on the NSW Electricity Price dataset with a forecast horizon of 48. We injected controlled perturbations into residual windows, including horizon-level shifts, zero-mean shape distortions, localized spikes, and a mixed setting containing all three types. Since the injected components are known, this setting allows direct comparison between the decomposed components and the component-level ground truth.

The perturbations are constructed as follows. For the **level-shift** setting, we add a constant offset to all future steps within the prediction horizon, which represents a persistent residual shift. For the **shape-distortion** setting, we add a zero-mean sinusoidal temporal pattern across the horizon, which changes the intra-horizon profile without changing the average level. For the **localized-spike** setting, we inject a small number of large positive or negative impulses at randomly selected horizon positions. For the **mixed** setting, we combine all three perturbation types in the same residual window. This gives us known ground-truth slow, shape, and spike components for each synthetic sample.

In the table below, **Synthetic setting** denotes the type of controlled perturbation injected into the residual window. **Intended effect** denotes the REIN channel that is expected to capture the injected perturbation. **Main validation metric** gives the quantitative criterion used to compare the decomposed component with the known ground-truth perturbation. **Result** reports the value of that metric, where higher values indicate better alignment.

The metrics are computed as follows. **Slow correlation** is the Pearson correlation between the decomposed slow scalar and the ground-truth level shift across synthetic samples. **Slow energy ratio** is the fraction of the decomposed residual energy assigned to the slow channel in the level-shift setting. **Shape correlation** is the Pearson correlation between the decomposed shape vector and the injected zero-mean shape pattern over all samples and horizon steps. **Shape energy ratio** is the fraction of decomposed residual energy assigned to the shape channel in the shape-distortion setting. **Spike precision**, **spike recall**, and **spike F1** compare the decomposed spike mask with the known injected spike locations.

Synthetic setting	Intended effect	Main validation metric	Result
Level shift	Slow channel	Slow correlation	1.000
Level shift	Slow channel	Slow energy ratio	1.000
Shape distortion	Shape channel	Shape correlation	0.987
Shape distortion	Shape channel	Shape energy ratio	0.971
Localized spike	Spike channel	Spike precision	1.000
Localized spike	Spike channel	Spike recall	1.000
Localized spike	Spike channel	Spike F1	1.000
Mixed effects	Slow channel	Slow correlation	0.999
Mixed effects	Shape channel	Shape correlation	0.963
Mixed effects	Spike channel	Spike F1	0.930

The results show that level-shift perturbations are assigned almost entirely to the slow channel, shape distortions are captured primarily by the shape channel, and localized spikes are accurately identified by the spike channel. In the mixed setting, the decomposition remains well aligned with the three intended temporal effects, with high slow correlation, high shape correlation, and strong spike detection performance.

These results provide controlled evidence that the slow/shape/spike decomposition corresponds to meaningful temporal residual patterns rather than arbitrary latent branches.

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Official Review of Submission16816 by Reviewer TZqL

Official Review by Reviewer TZqL 22 Jun 2026, 19:41 (modified: 29 Jul 2026, 04:37)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer TZqL

Revisions (/revisions?id=F76jwGjqz5)

Summary:

The authors introduce REIN (Residual Information from News), a model consisting in an LLM agent which derives actions from the news to modulate the residuals of a base forecaster through three modalities : persistent level, intra-day shape and extreme event spike. The LLM transforms the news input into several features along with its applicability horizon and a confidence level for its usage in the correction. At each time step t, the news which were published within a certain window around t are selected, embedded by the LLM into the feature space, aggregated and weighed according to their age and confidence level. This embedding is used by the model to control the 3 correction levels. The

correction is bounded to avoid degradation. The author prove the relevance of their method compared to strong baselines (deep time-series models, foundation models, LLM-based models) on 4 datasets including electricity price, demand, gas demand and traffic. They provide a counterfactual evaluation which shows the relevance of presence, content and alignment in the global performances of the model.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Overall, I found the paper well written. The objective and contributions are clearly exposed and the experiments show the strengths and limitations of the proposed approach. Residual correction of a base forecast using news seems relevant in many domains. The originality of the approach lies in the specific way the residual correction is done (as a weighed sum of the three components slow, shapel, spike where only the weights are new-dependent with the LLM prompted using a strict design to enhance interpretation), the temporal alignment of the news and the penalization of the training loss with a counterfactual penalty. I found the idea interesting and promising. The detailed prompts for the LLM agent is also useful. My only concern is that I had a hard time linking the modules from Figure 1 and the terms introduced in the formulas in Section 3.

Quality: 4: excellent
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:

- I suggest dropping the underscore z to indicate z-space normalization since the reader already expect the data to be normalized and the notation may mislead the reader into thinking that z is another variable.
- I think Figure 1 would benefit from being larger and including variable names. For example, in the DELTA STAGE, replace the node "SLOW RESIDUAL" with "SLOW RESIDUAL (r_{slow})". This would make it easier to link the graphical representation with the mathematical equations but I understand that this might not be possible due to space constraints and that it could overcrowd the figure. Maybe another solution could be to add a table in the Appendices to gather the notation used and their relation to Figure 1.
- In the JSON produced by the LLM, the term `renewable_surplus` may be too specific to the energy sector and does not need to be since, as I understood is, it acts as the opposite of `tightness`. I don't know if this is a relevant concern or if all that matters is the way the LLM is told to interpret this slot as in the Traffic example `renewable_surplus = use this slot as mobility relief and alternate-capacity availability that reduces road counts or congestion, negative when relief is weak and road usage is pushed higher`".
- In Eq. (6), I am not sure what the overline in $\overline{\hat{g}_{\text{splike}}}$ and $\overline{\mathbb{S}}$ refers to.
- In Appendix K, adding plots of the base model prediction agains the target would help understand what the correction is trying to correct. For example in the Electricity Load example, the correction exhibits a clear daily pattern. It this because the base forecast is flat and the correction recovers the seasonality of the Load which was not captured by the base model ? Adding the base forecast and target plots would help answer this type of questions.
- Figure in Appendix K would be cleaner if they did not include the name of the .csv file below each plot. They are very small but, since I had to zoom in to see the figures, I could also see the names the the .csv files.
- Although this is not necessary for the relevance of the work, I am wondering how the approach would perform if we test the model on the COVID period (without seeing it on the training data) as lockdowns had a large influence on energy demand and traffic.

Limitations:
Yes

Rating: 5: Accept: Technically solid paper, with high potential value on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
None

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

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Rebuttal by Authors

Rebuttal

by Authors (Qingwen Zeng (/profile?id=~Qingwen_Zeng1), Huaming Chen (/profile?id=~Huaming_Chen1), Zhaoge Bi (/profile?id=~Zhaoge_Bi1), Linghan Huang (/profile?id=~Linghan_Huang1), +1 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission16816/Authors))

28 Jul 2026, 21:51 (modified: 28 Jul 2026, 23:35)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=4n4k8NemXW)

Rebuttal:

We sincerely thank the reviewer for raising these insightful questions, which encouraged us to further clarify the motivation and limitations of REIN.

Response to Question 1

We agree that using the subscript (z) may be unnecessary and could potentially confuse readers by suggesting that it denotes a separate variable. In the revised manuscript, we will remove this notation and clarify in the text that the corresponding quantities are normalized in the standard (z)-space.

Response to Question 2

It is true that explicitly linking the nodes in Figure 1 to the mathematical notation would improve readability and make the connection between the graphical illustration and the equations clearer. We will enlarge Figure 1 where possible and revise the node labels to include the corresponding variable names, such as “SLOW RESIDUAL ((r_{\mathrm{slow}}))”. If space constraints make it difficult to include all notation directly in the figure, we will also add an appendix table summarizing the main symbols and their correspondence to the components in Figure 1. This should help readers connect the visual pipeline with the mathematical formulation more easily.

Response to Question 3

In REIN, the term `renewable_surplus` is intended to represent a relieving or offsetting factor opposite to `tightness`, rather than a concept limited to renewable energy. For example, in the traffic setting, this slot is interpreted as mobility relief or alternate-capacity availability that can reduce road counts or congestion, while negative values indicate weak relief or conditions that push traffic usage higher. We will revise the manuscript to clarify that the semantic meaning of this field is adapted to each domain through the LLM prompt.

Response to Question 4

The overline in $\hat{g}_{\mathrm{spike}}$ and $\bar{S}$ denotes a normalized or averaged reference quantity used in the corresponding modulation term. We recognize that this was not sufficiently clear in the original manuscript. We will revise the explanation around the equation so that all notation is defined before use.

Response to Question 5 and 6

We agree that adding plots of the base model prediction together with the target series would make the role of the correction easier to understand. We will add additional plots in Appendix K showing the base forecast, the corrected forecast, and the ground-truth target. These plots will help clarify what the REIN correction is trying to correct in each case. And including `.csv` file names below each plot makes the appendix figures less clean, especially when readers zoom in. We will remove these file names from the plots and revise the figure captions to describe the selected examples in a cleaner and more readable way.

Response to Question 7

This is a very interesting suggestion. Evaluating the model during the COVID period would be a meaningful stress test, since lockdowns caused substantial distribution shifts in both energy demand and traffic patterns. Although this analysis is beyond the main experimental scope of the current manuscript, it is highly relevant for assessing robustness under extreme external events. We will add this point to the discussion as an important direction for future work. In particular, such an evaluation would test whether REIN can use news-derived external signals to improve forecasting under rare and disruptive conditions that are not well represented in the training data.

Add: **Official Comment** **Author AC Confidential Comment**

Official Review of Submission16816 by Reviewer uvzL

Official Review by Reviewer uvzL 21 Jun 2026, 11:38 (modified: 29 Jul 2026, 04:37)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer uvzL

Revisions (/revisions?id=k7ZuCmfDsq)

Summary:
This paper introduces REIN, a new time-series forecasting framework that integrates textual news by treating it as a bounded modifier rather than an unconstrained feature. By using an LLM to distill raw news into compact regime signals, the model selectively scales decomposed residual channels on top of a frozen numerical base forecaster. This design ensures that the influence of external events is capped and reverts to a reliable numerical baseline when no actionable news is present.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strength

1. It provides a architectural guarantee that limits how much textual data can perturb the base forecast, preventing runaway errors in operational settings.
2. The framework isolates the impact of text by decomposing forecast corrections into distinct channels.
3. It introduces a counterfactual testing protocol to prove that accuracy gains stem from time-aligned news rather than mere text availability.

Weakness

1. REIN fails to outperform numerical baselines on the Traffic dataset.
2. The paper fails to provide a component-wise ablation study to quantify the individual contributions of the three decomposed residual channels.
3. Relying on a LLM to process raw news adds computational overhead and architectural complexity to the pipeline.

Quality: 2: not good
Clarity: 3: good
Significance: 2: not good
Originality: 3: good
Questions:
Could the authors provide a component-wise ablation study of the three residual channels?

Limitations:
Yes

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
N/A

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

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Rebuttal by Authors

Rebuttal

by Authors (Qingwen Zeng (/profile?id=~Qingwen_Zeng1), Huaming Chen (/profile?id=~Huaming_Chen1), Zhaoge Bi (/profile?id=~Zhaoge_Bi1), Linghan Huang (/profile?id=~Linghan_Huang1), +1 more (/group/edit?id=NeurIPS.cc/2026/Conference/Submission16816/Authors))

28 Jul 2026, 21:57 (modified: 28 Jul 2026, 23:35)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Revisions (/revisions?id=NYyQJ89D9A)

Rebuttal:

Response to Question 1 and Weakness 2

We thank the reviewer for highlighting this issue, it has helped us make the revised version clearer and more rigorous. A component-wise ablation study is the most direct way to evaluate whether the slow/shape/spike decomposition is useful beyond simply increasing model capacity. We haved added ablations where each residual channel is removed while keeping the rest of the training and regime-bank pipeline unchanged. Specifically, we will evaluate `w/o active threshold`, `w/o slow`, `w/o shape`, and `w/o spike`, and compare them with both the full REIN model and a single-head residual variant. The results will be updated soon.

These ablations will also help interpret the role of each channel across datasets. For example, we expect the slow channel to be more important for persistent demand or gas-demand shifts, the shape channel to be more relevant for traffic and holiday-related intra-day changes, and the spike channel to be particularly useful for heavy-tailed price movements or incident-driven corrections. If the ablation results do not support this interpretation, we will revise the discussion accordingly and avoid overclaiming the semantics of the channels.

Response to Weakness 1

A likely reason is that the traffic target is highly local and sensor-dependent, whereas the available GDELT-derived news stream is broader and does not always capture the specific road incidents, local closures, commuter shifts, or station-level disruptions that drive hourly traffic counts. In this setting, the bounded design helps prevent large degradation, but the potential gain from weakly matched news is naturally limited.

We will revise the discussion to make this limitation explicit. We also believe that the Traffic result is informative because it shows that REIN does not automatically improve a forecast simply by adding text. Instead, REIN is most effective when the news corpus contains timely and actionable information aligned with the target series. We will strengthen the active/blank/permuted analysis for Traffic so that readers can better see where aligned news helps and where corpus coverage is insufficient.

At the same time, the Traffic results still support the value of REIN. Compared with ChatTime under the same retrieved news corpus (experiments on other horizons are not finished yet), REIN achieves substantially better performance, indicating that the proposed bounded residual-modulation mechanism is more effective than directly incorporating text in this noisy and weakly aligned setting. More importantly, the Traffic result demonstrates a useful boundary condition of our approach: REIN improves forecasting when the external corpus contains timely and actionable information, but it does not guarantee improvement when the corpus lacks sufficient local relevance. We will make this limitation clear in the revised manuscript.

Dataset	Horizon	ChatTime's MSE	ChatTime's MAE	REIN's MSE	REIN's MAE
Electricity price	48	564044.4824	103.4364719	561709.8272	107.3152611
Electricity price	96	885512.6037	142.81023	580792.2357	120.3588648
Electricity demand	48	1503324.343	889.2976611	503629.2308	469.8057475
Electricity demand	96	2019168.523	1079.905995	658304.6118	554.1322286
Traffic	48	1003876539	22486.90546	327345499.6	12036.87399
Traffic	96	1944465970	31551.93068	365076237.5	13098.16765
Gas Demand	48	5.83E+13	5859254.306	3.02152E+13	4278464.786
Gas Demand	96	6.84E+13	6548942.089	3.74725E+13	4997697.726

Response to Weakness 3

We thank the reviewer for raising this practical concern. REIN has a relatively small GPU-memory footprint. The LLM is only used offline to construct cached regime records and is not loaded during forecasting. The online delta module contains only about 2.0M--2.25M parameters across horizons 48--720, corresponding to 7.6--8.6 MB of FP32 weights and about 30--34 MB of Adam training states. This explains why REIN is substantially more memory-efficient than LLM-in-the-loop text forecasting baselines such as ChatTime.

A practical advantage of REIN is that, once constructed, the regime bank can be stored offline as a persistent and model-independent representation of the external information. Once the regime bank is established, the online forecaster only retrieves compact cached features, including numerical regime vectors, topic masses, embeddings, and relevance mass. As a result, the forecasting-time cost remains close to that of an ordinary residual forecasting model rather than an LLM-in-the-loop system. This decouples the costly processes of news collection, filtering, temporal alignment, and semantic refinement from downstream forecasting, allowing the same regime records to be reused by different predictors, fusion mechanisms, or decision algorithms without repeatedly processing the raw news. Such modularity also simplifies experimentation, since numerical models can be replaced or compared while the external-information pipeline remains fixed, and it improves reproducibility, caching efficiency, inference latency, and auditability. By contrast, ChatTime and Time-VLM integrate textual or multimodal information more tightly into their model-specific tokenisation, representation, and fusion pipelines, making those representations less directly reusable outside the original architecture. From an engineering perspective, REIN therefore offers a more convenient and extensible interface between external information and forecasting algorithms.

In addition, our newly added comparison with the news-aware baseline ChatTime further illustrates this point. REIN outperforms ChatTime in most experimental settings while requiring fewer computational resources. In particular, ChatTime could not be successfully run for the longer horizons of 336 and 720 on our RTX 5090 platform due to GPU memory constraints. This suggests that REIN provides a more computationally efficient way to incorporate textual information, especially for long-horizon forecasting.

Furthermore, although REIN contains several explicit components, related multimodal forecasting models such as ChatTime and Time-VLM are not necessarily simpler and, in some respects, involve more extensive design and/or fine-tuning. ChatTime includes time-series normalisation and discretisation, vocabulary expansion, specialised tokenisation and de-tokenisation, inverse scaling, continuous pre-training on a large time-series corpus, and multi-task instruction fine-tuning for forecasting and question answering. Time-VLM similarly uses three parallel learners for temporal, visual, and textual information, together with a historical memory bank, frequency- and periodicity-aware image generation, frozen VLM encoders, cross-modal attention, and gated multimodal fusion. By comparison, REIN retains a frozen numerical forecasting backbone and adds a structured, bounded residual correction conditioned on temporally aligned news. Its additional modules therefore serve a relatively focused and controlled purpose, rather than constructing an entirely new multimodal foundation model or representation space. From this perspective, REIN appears broadly comparable to these models in architectural complexity and may require a more limited degree of model adaptation and training.

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